Assignment5 – Markov Decision Procedures and Decision Trees

Given: May 20 Due: May 25

Problem 5.1 (HMMs in Python)

Implement filtering, prediction and smoothing for HMMs in Python by completing the implementation of `hmm.py` at `https://kwarc.info/teaching/AI/resources/AI2/hmm/`.

Hint: This problem uses `numpy`, which is a Python library for working with arrays/matrices. If you have never worked with `numpy` before, you can find many high-quality introductions online. Due to its popularity and frequent use for machine learning etc., it is definitely worth getting to know `numpy`. That being said, you only need very few and basic numpy functions for this assignment, which you should be able to find without problems (e.g. searching for *numpy matrix multiplication*).

Solution: See `https://kwarc.info/teaching/AI/resources/AI2/hmm/`.

Problem 5.2 (Value Iteration for Navigation)

Implement value iteration for an agent navigating worlds like the 4x3 world from the lecture notes. The agent has four possible actions: *right*, *up*, *left*, *down*. The probability of actually moving in the intended direction is p and the probability of moving in one of the orthogonal directions is $\frac{1-p}{2}$ respectively. For example, if $p = 0.8$ and the chosen action is *up*, the agent will actually move up with a probability of $p = 0.8$ and will move left and right with a probability of 0.1 each. If the agent ends up moving in a direction that has no free adjacent square, it will remain on its current square instead. For example, if the agent is on square (0,0) with the action *up*, it will end up on square (0,1) with a probability of p, on square (1,0) with a probability of $\frac{1-p}{2}$ and on square (0,0) with a probability of $\frac{1-p}{2}$.

(0, 2) -0.040 → 0.647	(1, 2) -0.040 → 0.753	(2, 2) -0.040 → 0.855	(3, 2) 1.000 T 1.000
(0, 1) -0.040 ↑ 0.557	(1, 1) W	(2, 1) -0.040 ↑ 0.569	(3, 1) -1.000 T -1.000
(0, 0) -0.040 ↑ 0.465	(1, 0) -0.040 ← 0.386	(2, 0) -0.040 ↑ 0.451	(3, 0) -0.040 ← 0.230

Results for 4x3 world with $p = 0.8$, $\gamma = 0.95$, $\varepsilon = 0.001$.

A skeleton implementation with technical instructions can be found at `https://kwarc.info/teaching/AI/resources/AI2/mdp/`. It also allows the visualization of the computed utilities and policy (see figure above): Each square is annotated with the coordinates, the reward, the computed policy and the computed utility. Walls and terminal nodes don't have a policy and are marked with W and T respectively.

Hint: You will also have to compute a policy based on the utilities obtained from value iteration. For that, you should pick the actions that maximize the expected utility. A common mistake is the assumption that the best policy is always to go in the direction of the square with the maximal utility.

Solution: See `https://kwarc.info/teaching/AI/resources/AI2/mdp/`

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instead of tracking one state
the agent tracks a belief state.

Partially Observable Markov Decision Process

Problem 5.3 (POMDPs)

An agent is moving on an infinite discrete line. We represent its position as an integer.

z

Initially, it is in position 4 with 60% probability or 5 with 40% probability. Its goal is to reach position 0.

P(s=4)=0.6
P(s=5)=0.4

-1 0 1

Its actions are to move left or right 1 step or to do nothing. Making a step succeeds 75% of the time; the remaining times, the agent does not move. Doing nothing always succeeds, i.e., does not change the state.

After each action the agent measures which state it is in. This measurement is correct 50% of the time and one off in either direction 25% of the time.

We want to model this as POMDP.

Hint: We often have to give probability distributions, e.g., over S. Such a distribution is a function $p : S \to [0;1]$. Often $p(s') = 0$ for all but finitely many values, e.g., $p(s_0) = p(s_1) = 0.5$ and $p(s) = 0$ otherwise. Such a function p is a set of

⟨S,A,I,T,R,E,O⟩
S state space
A actions
I initial belief
T transition model
R reward function
E observation space
O observation model

pairs $(s', r) \in S \times [0;1]$ where $r = p(s')$. In that case, we can write p efficiently by listing only the non-zero values, e.g., $p = \{(s_0, 0.5), (s_1, 0.5)\}$. Technically, we have $p = \{(s_0, 0.5), (s_1, 0.5)\} \cup \{(s, 0) : s \in S \setminus \{s_0, s_1\}\}$, but we can clarify that with a comment that all omitted values are 0. That is a convenient way to write a concrete distribution over a large domain where only a few values are non-zero.

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1. Give the state space S and the set A of actions.

true stare may be hidden but it exists

Solution: $S = \mathbb{Z}$, and $A = \{-1, 0, 1\}$ (same for all states).

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2. Give the transition model $T(s, a)$ as a probability distribution over states.

Solution: $T(n, 1) = \{(n+1, 0.75), (n, 0.25)\}$, and $T(n, 0) = \{(n, 1)\}$, and $T(n, -1) = \{(n-1, 0.75), (n, 0.25)\}$.

Result	Probability
moves to (n+1)	0.75
stays at (n)	0.25

doing nothing always succ.

3. Give the initial state I and define an appropriate reward function R.

The goal is to reach position 0.

Solution: $I = \{(4, 0.6), (5, 0.4)\}$. For example, $R(0) = 1$ and $R(s) = -0.1$ otherwise.

ie. if in 0 reward 1 else penalty -0.1

Observation	Probability
correct state	0.5
one too high	0.25
one too low	0.25

4. Give the domain E of observations and the observation model $O(s, e)$.

observations are noisy

Solution: Here the observations simply measure the state we're in. So $E = S = \mathbb{Z}$.
$O(s, e)$ is the probability of measuring e in state s. So $O = \{(s, s+1, 0.25), (s, s, 0.5), (s, s-1, 0.25)\}$.

All other observations have probability 0.

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5. The agent initially moves 1 step to the right in belief state $b_0 = I$. What are the next belief state b_1 (before taking the next measurement) and its expected reward $\rho(b_1)$ and the distribution of the next observation E_1?

Solution: After performing $a = +1$, we have $b_1 = \{(4, 0.6 \cdot 0.25), (5, 0.6 \cdot 0.75 + 0.4 \cdot 0.25), (6, 0.4 \cdot 0.75)\} = \{(4, 0.15), (5, 0.55), (6, 0.3)\}$.
We obtain its expected reward by summing over the states: $\rho(b_1) = R(4) \cdot 0.15 + R(5) \cdot 0.55 + R(6) \cdot 0.3 = -0.1$.

all possible states are non-goal states

We obtain the probability distribution for E_1 by multiplying $b_1(s')$ and $O(s', e)$ and summing over s' for each possible observation e. The 3 non-zero entries in b_1 and the 3 non-zero entries in $O(s, e)$ yield 9 terms that add up to the

The agent may actually be at state 5
ie. Transition model T (truth)
but measure 4 ie.
observation model O (sees)

State 4: 0.15
State 5: 0.45+0.1=0.55
State 6: 0.3
So: b1 ={(4,0.15),(5,0.55),(6,0.3)}

Check:
0.15+0.55+0.3=1

New state	Calculation	Probability
5	(0.6 \cdot 0.75)	0.45
4	(0.6 \cdot 0.25)	0.15

at 4

New state	Calculation	Probability
6	(0.4 \cdot 0.75)	0.3
5	(0.4 \cdot 0.25)	0.1

at 5

$$P(e) = \sum_s b_1(s) \cdot O(s, e)$$

True state 4 → obs 3

TS: 5 → obs 4

TS 6 → obs 5

etc.

following distribution: $\{(3, 0.15 \cdot 0.25), (4, 0.15 \cdot 0.5 + 0.55 \cdot 0.25), (5, 0.15 \cdot 0.25 + 0.55 \cdot 0.5 + 0.3 \cdot 0.25), (6, 0.55 \cdot 0.25 + 0.3 \cdot 0.5), (7, 0.3 \cdot 0.25)\}$.

Bayesian update

$b'(s) =$

$$\frac{b_1(s) \cdot O(s, e)}{\sum_{s'} b_1(s') \cdot O(s', e)}$$

$P(s=4 \mid e=4) = \frac{0.075}{0.2125} \approx 0.35$

$= \frac{0.1375}{0.2125} \approx 0.65$

6. After the move from the previous question, we measure that we are in state 4. What is the updated belief state?

Solution: We can reuse the terms $0.15 \cdot 0.5$ (stemming from $s' = 4$) and $0.55 \cdot 0.25$ (stemming from $s' = 5$) for the observation 4. This yields the vector $\{(4, 0.075), (5, 0.1375)\}$, which normalizes to approximately $\{(4, 0.35), (5, 0.65)\}$.

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7. Give the corresponding MDP $\langle S', A', I', T', R' \rangle$. For T', don't give the precise definition and simply explain how to create it.

Solution: States are belief states: S' is the set of probability distributions over $\mathbb{Z}$, i.e., functions $b : \mathbb{Z} \to [0; 1]$ where all values sum to 1.
Actions are the same as for the POMDP. Technically, we'd have to say $A(b) = \bigcup_{s \in \mathbb{Z}, b(s) \neq 0} A(s)$ to handle the fact not every action might be allowed in every state. But our particular POMDP anyway allows the same actions in all states. So we simply have $A' = A = \{-1, 0, 1\}$.
The initial state is the same as that of the POMDP: $I' = I$.
The reward function calculates the expected reward in a belief state (like $\rho(b)$ above): $R'(b) = \Sigma_{s \in \mathbb{Z}} R(s) \cdot b(s)$.
The transition model is the hard part. $T(b, a)$ is a probability distribution over the belief states b' that arise from executing action a in belief state b. We obtain it as follows:

In the original POMDP: T(s,a) moves from a real state to another real state.

In the belief-MDP: T' (b,a) moves from one belief state to another belief state.

1. Executing action a in b directly determines a new belief state b_a by summing over all states (as done for a special case in the previous question): $b_a(s') = \Sigma_s T(s, a)(s') \cdot b(s)$ where $T(s, a)$ returns the successor state as a probability distribution and $T(s, a)(s')$ is the probability of getting successor state s'.

   The probability of ending in s' is the sum of all ways to come from earlier states s.

2. Then we make an observation e, which we use to update our belief state to b'. Here $b_a(s')$ is the prior probability of being in s'; $O(s', e)$ is the conditional probability of observing e when in s'; multiplying them yields the probability $P(s', e)$ of being in state s' and observing e.

   We can do 2 things with these values:

   - Summing $P(s', e)$ over s' yields the probability $P(e)$ of observing e (as done for a special case in the previous question).
   - Fixing e and normalizing the resulting s'-indexed vector $P(s', e)$ yields the updated probability of being in s' after observing e. Call that vector $b'_{a,e}$.

From belief b, after action a, the next belief is b'_a,e with probability P(e).

3. Every possible observation e yields a new belief state $b'_{a,e}$ with probability $P(e)$ (as computed above). So essentially $T'(b,a)$ assigns the probability $P(e)$ to the belief state $b'_{a,e}$. But it may happen that some values of e yield the same belief state $b'_{a,e}$. So $T'(b,a)(b')$ technically arises as the sum of all $P(e)$ that yield the belief state b'.

Problem 5.4 (Sunbathing)

Eight people go sunbathing. They are categorized by the attributes Hair and Lotion and the result of whether they got sunburned.

Name	Hair	Lotion	Result: Sunburned
Sarah	Light	No	Yes
Dana	Light	Yes	No
Alex	Dark	Yes	No
Annie	Light	No	Yes
Julie	Light	No	No
Pete	Dark	No	No
John	Dark	No	No
Ruth	Light	No	No

1. Which quantity does the information theoretic decision tree learning algorithm use to pick the attribute to split on?

Solution: Information gain.

Choose the attribute that reduces uncertainty the most.
ie. Before splitting, we are uncertain about who gets sunburned.
After splitting by an attribute, we hope the groups become more pure.
A group is pure if all examples in it have the same label.

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2. Compute that quantity for the attributes Hair and Lotion. (Simplify as much as you can without computing logarithms.)

Solution:

(yes,no)

$$E_0 := I(\langle \tfrac{2}{8}, \tfrac{6}{8} \rangle) = -\tfrac{2}{8}\log_2(\tfrac{2}{8}) - \tfrac{6}{8}\log_2(\tfrac{6}{8}) \approx 0.81$$

light hair= 2 yes, 3 nos

$$\text{Gain}(Hair) = E_0 - \underbrace{\tfrac{5}{8} I(\langle \tfrac{2}{5}, \tfrac{3}{5} \rangle)}_{Light} - \underbrace{\tfrac{3}{8} I(\langle 0, 1 \rangle)}_{Dark} \approx 0.20$$

no lotion 2 yes, 4 no

$$\text{Gain}(Lotion) = E_0 - \underbrace{\tfrac{2}{8} I(\langle 0, 1 \rangle)}_{Yes} - \underbrace{\tfrac{6}{8} I(\langle \tfrac{2}{6}, \tfrac{4}{6} \rangle)}_{No} \approx 0.12$$

Entropy is undefined for 0. If we were to continue simplifying, we'd use $0 \cdot \log_2 0 = 0$.

3. Assuming the logarithms are computed, how does the algorithm pick the attribute?

Solution: It picks the one with the highest information gain (in this case Hair).

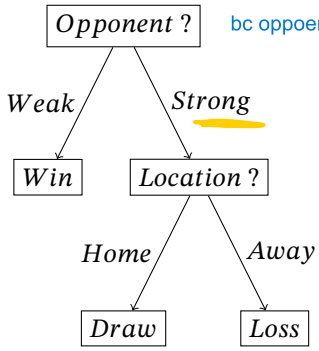
Problem 5.5 (Decision Trees)

You observe the values below for 6 different football games of your favorite team. You want to construct a decision tree that predicts the result.

#	Day	Weather	Location	Opponent	Result
1	Monday	Rainy	Home	Weak	Win
2	Monday	Sunny	Home	Weak	Win
3	Friday	Rainy	Away	Strong	Loss
4	Sunday	Sunny	Home	Weak	Win
5	Friday	Cloudy	Home	Strong	Draw
6	Sunday	Sunny	Home	Strong	Draw

1. Assume you choose attributes in the order $Opponent, Location, Weather, Day$.
Give the resulting decision tree.

Solution: The tree is



bc oppoent is PURE

after splitting by Opponent and Location, all branches are already pure.

A decision tree stops splitting when the examples in a node all have the same class.

2. Without using the above observations, give the formula for the information gain of the attribute $Opponent$.

Solution: $\text{Gain}(Opponent) = I(P(Result)) - P(Opponent = Strong) \cdot$

I is entropy

Entropy measures how mixed the outcomes are.
If the results are evenly split (e.g., Win/Loss/Draw), entropy is high.
If the results are mostly one class, entropy is low.

information gain definition:

Start with the entropy of the full dataset.

Subtract the weighted entropies of the subsets created by splitting on “Opponent”.

$I(P(Result \mid Opponent = Strong)) - P(Opponent = Weak) \cdot I(P(Result \mid Opponent = Weak))$

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3. Using the above observations, give the results of
   - $I(P(Result)) =$
   - $P(Result = Loss \mid Opponent = Strong) =$

   You do not have to compute irrational logarithms.

Solution: $I(P(Result)) = -1/2 \log_2 1/2 - 1/3 \log_2 1/3 - 1/6 \log_2 1/6$ and $P(Result = Loss \mid Opponent = Strong) = 1/3$.

P(win)= 3/6 = 1/2
p(draw) = 2/6 =1/3
p(loss =)1/6